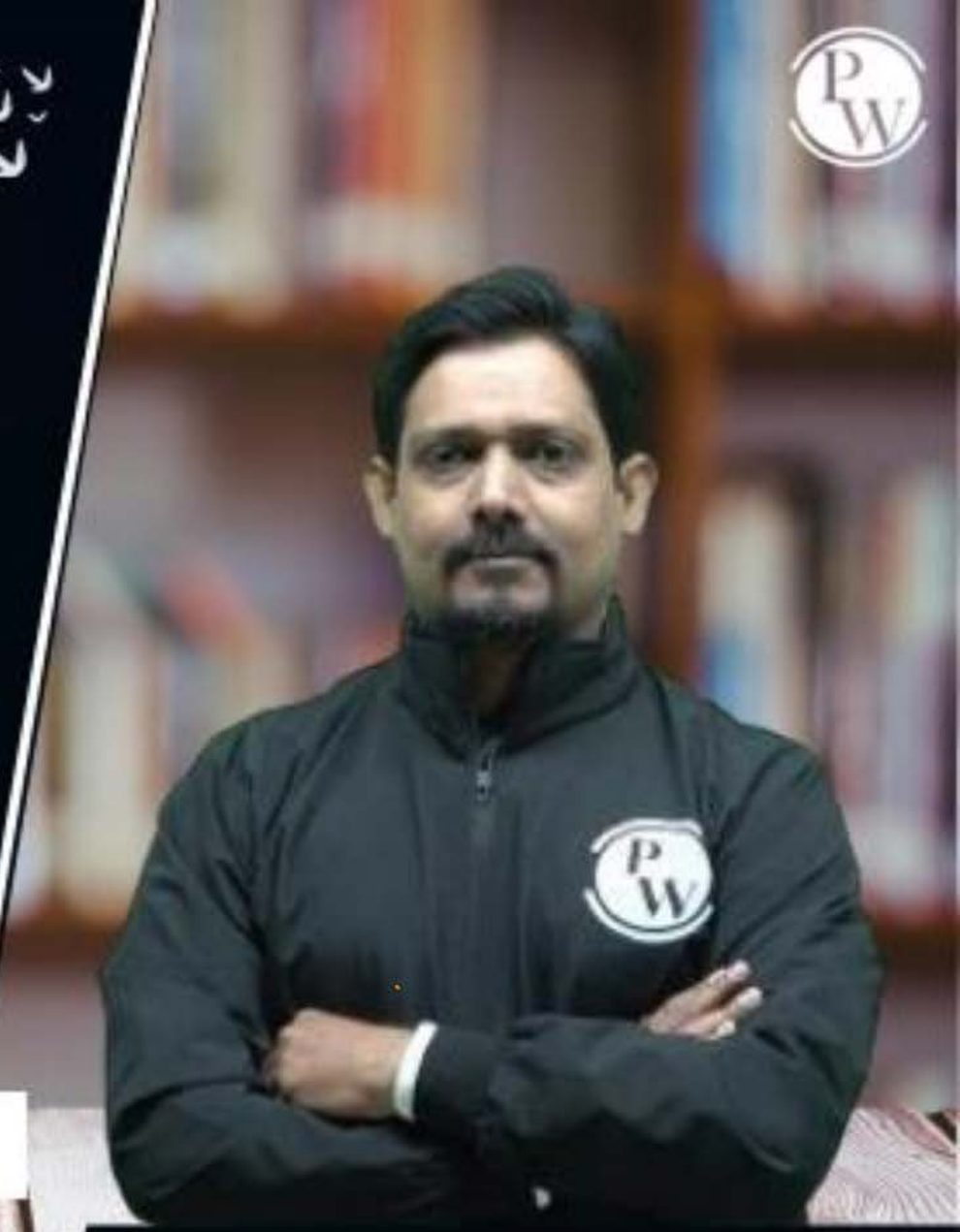
Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 08



By- Abhishek Sir

Recap of Previous Lecture



Topic

Topic

Topic

Topic

Topic

AVL with minimum No. of Node with height

Heap, CBT

Heap Insertion

Slide

Topics to be Covered



- Topic — Heap Deletion
- Topic — Adjust ✓
- Topic — Heapify
- Topic — Counting
- Topic

Slide



Topic : Tree

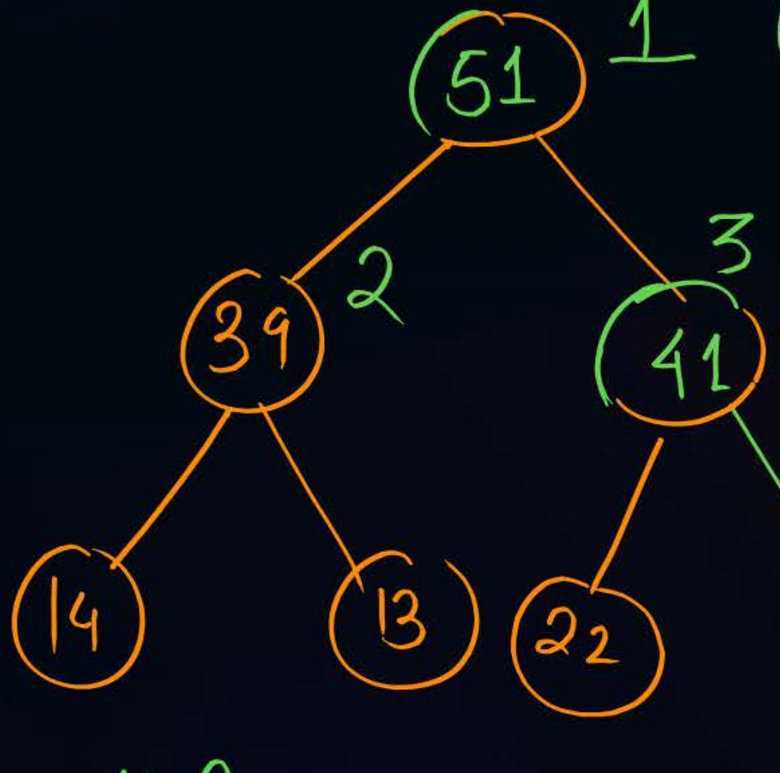
Heap Deletion
Adjust
Build heap
Counting

Slide



Topic : Heap

Max heap



Algorithm Insert(a, n, x) {

a[n] = x; i = n;

while (i > 1 and a[⌊i/2⌋] < x) {

Swap(a[i], a[⌊i/2⌋])

i = ⌊i/2⌋;

}

}

of Swap

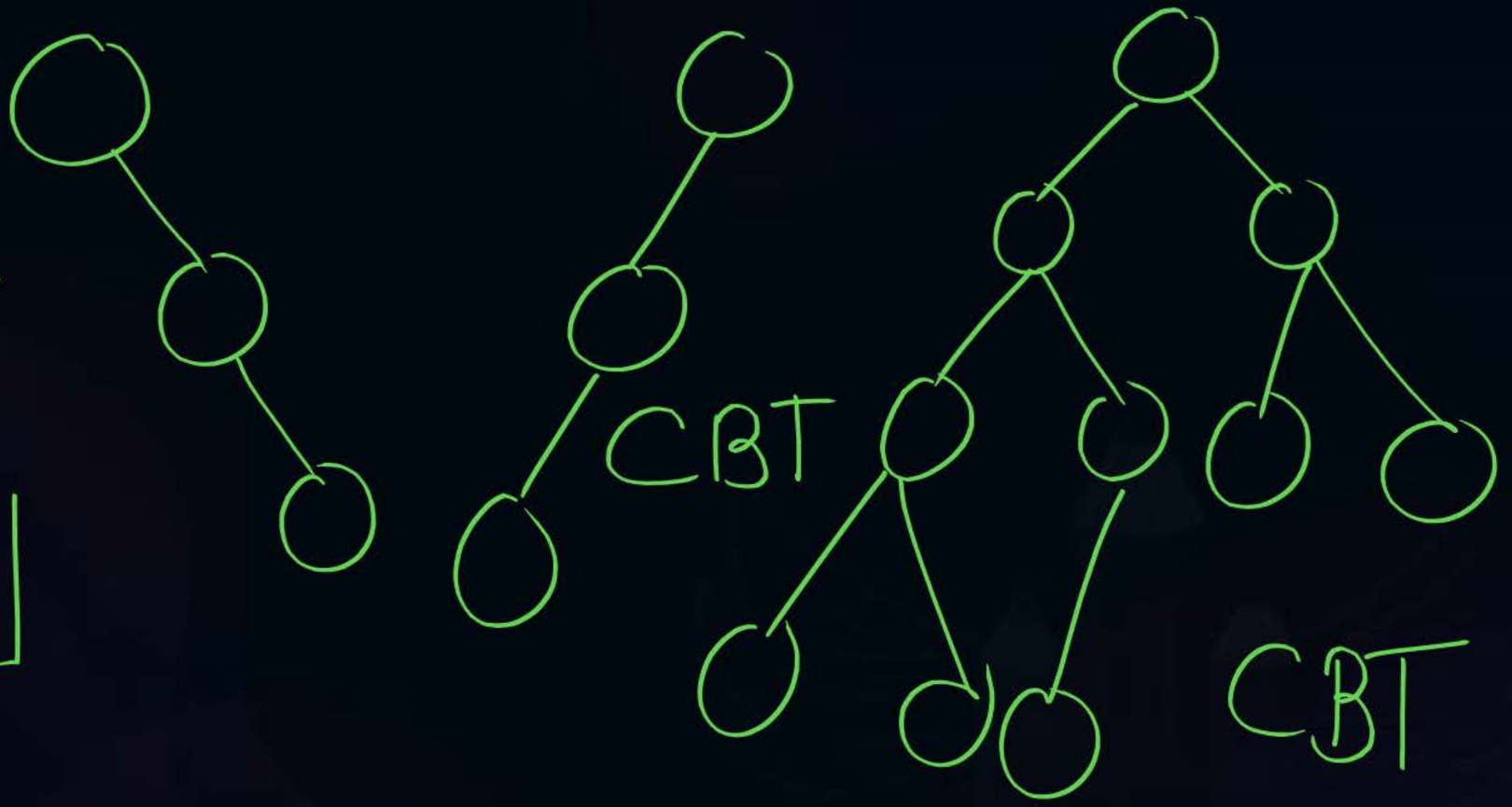
O(h)

= O(log₂ n)

Slide



Topic : Heap



$\lfloor \log_2 n \rfloor$

Slide



Topic : Heap

CBT height $\lfloor \log_2 n \rfloor$

Insert is $O(h)$

$O(\log_2 n)$

Topic : Heap

Delete

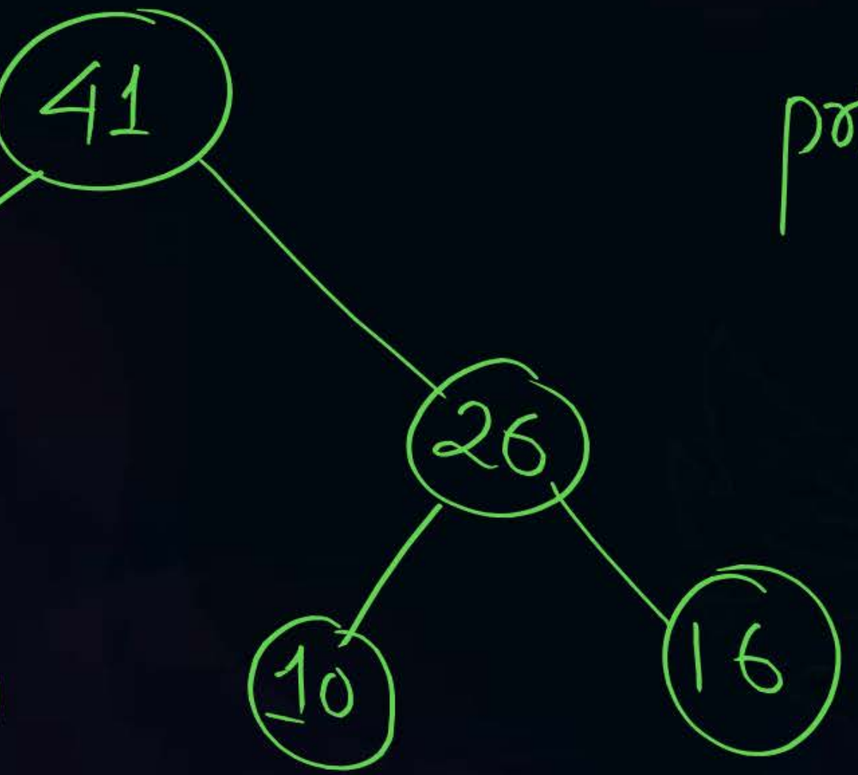
Max heap ✓

maximum

Min Heap

minimum

Delek



Insert

Adjust ←

priority queue

Slide



Topic : Heap

1. Delete Root

2. Copy Last element as Root

if only one element ← violation

violate Heap property ← Heap

then we call Adjust at index i where violation is happening.

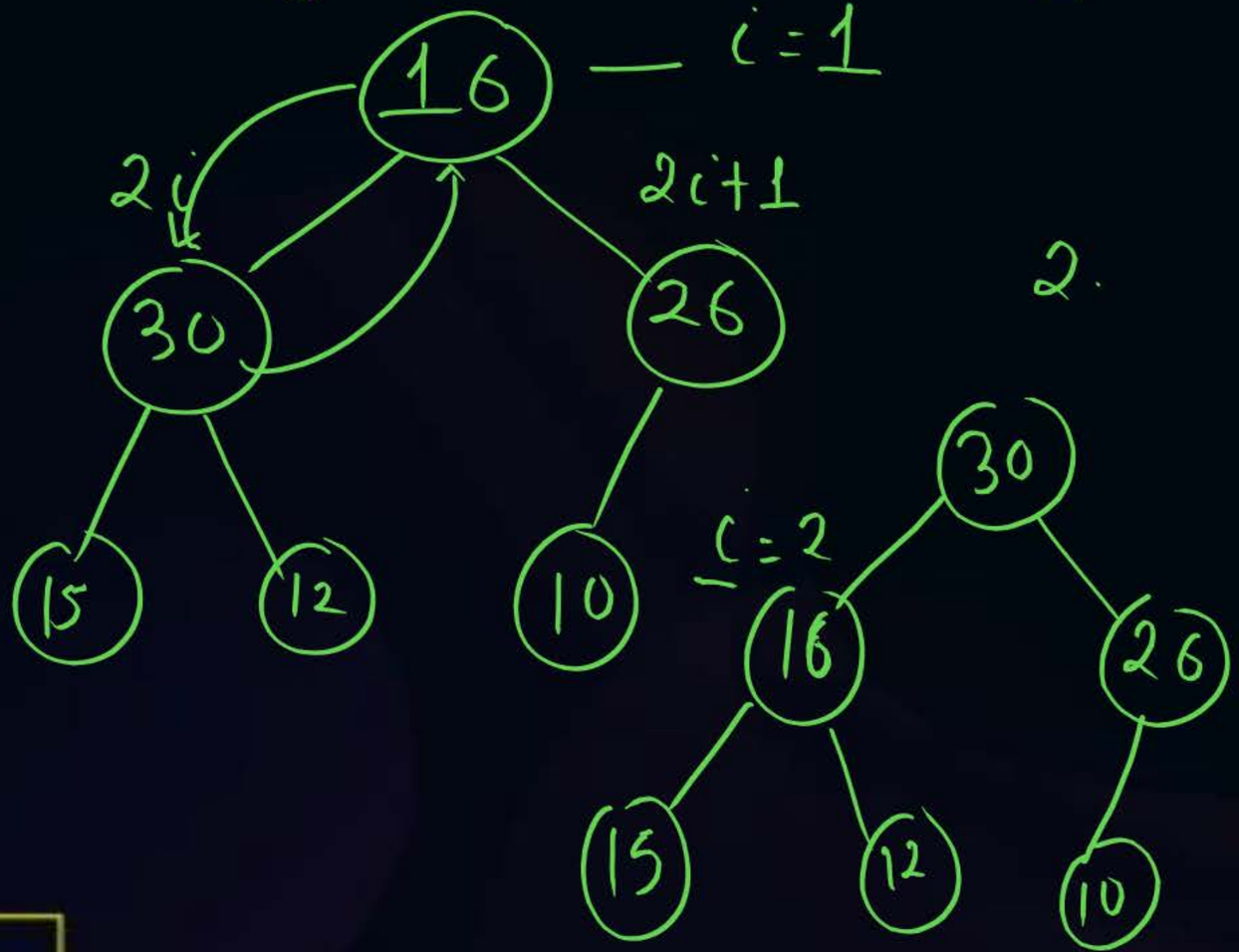


Slide



Topic : Heap

Adjust = Rearrange

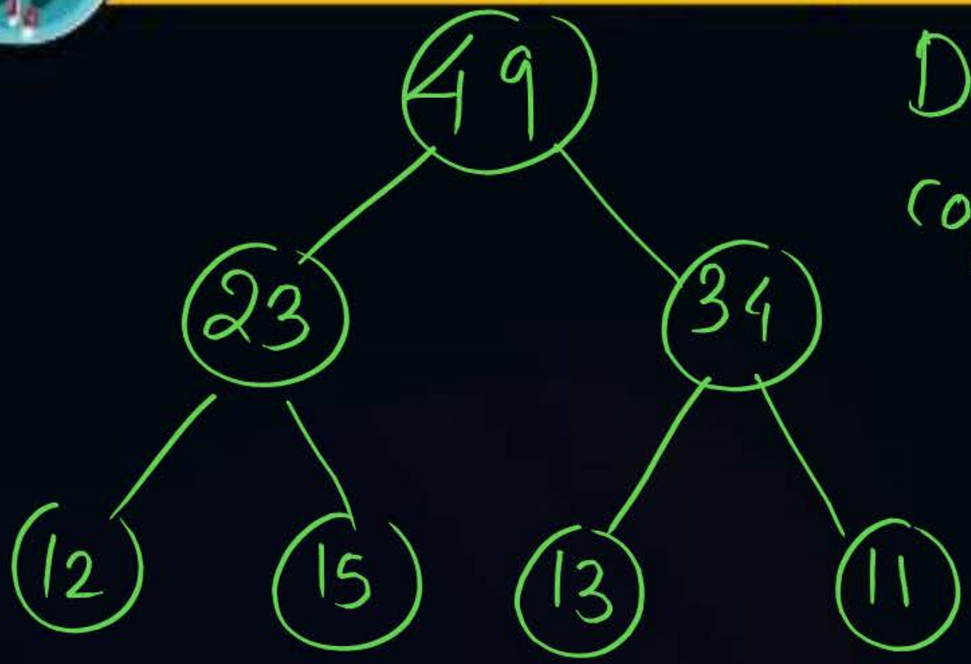


1. Compare left & Right child to find maximum
2. if max child value is greater than Node at index i then swap
3. update i to max child index

Slide

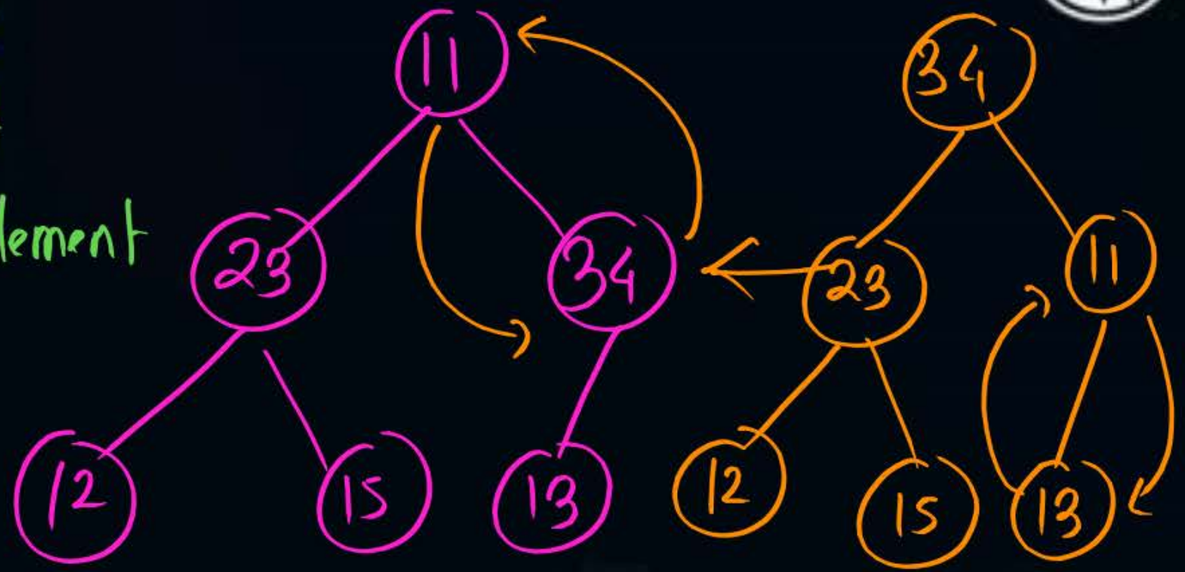


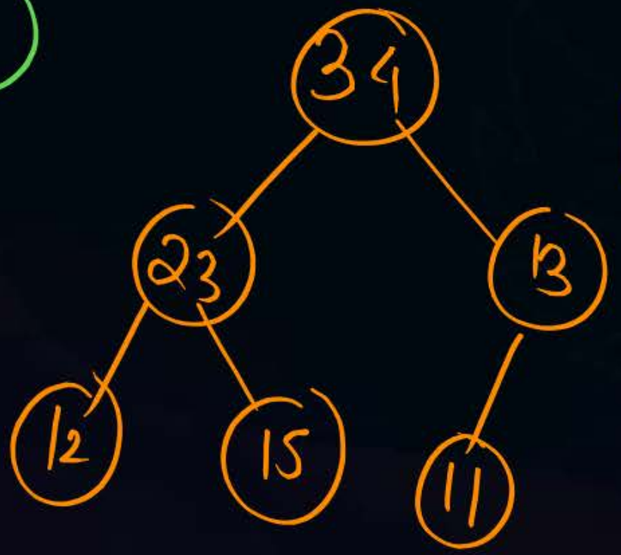
Topic : Heap



Delete Root

copy Last element





Max Heap

Slide



Topic : Heap



Delete and after Adjust

element at Array inde 4 is

a[4] = ?

Slide



Topic : Heap



4th Index _16

Slide



Topic : Heap

if heap is represented by {25,14, 16, 13, 10,8 ,12}

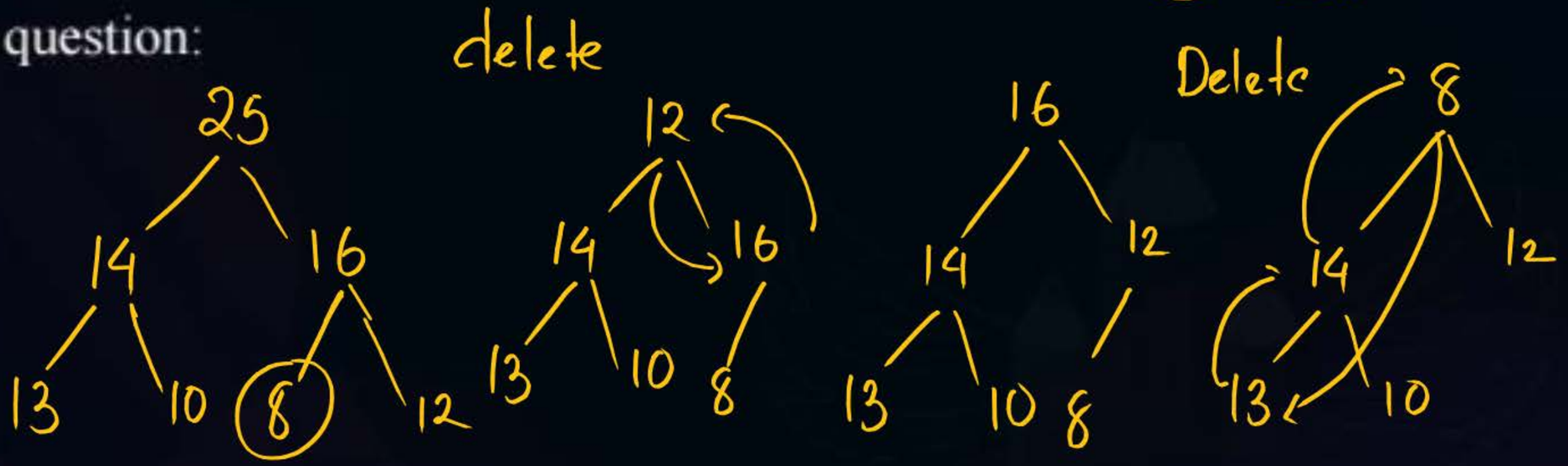
What is the content of the array after two delete operations on the correct answer to the previous question:

(a) {14, 13, 12, 10, 8}

(b) {14, 12, 13, 8, 10}

(c) {14, 13, 8, 12, 10}

(d) {14, 13, 12, 8, 10}



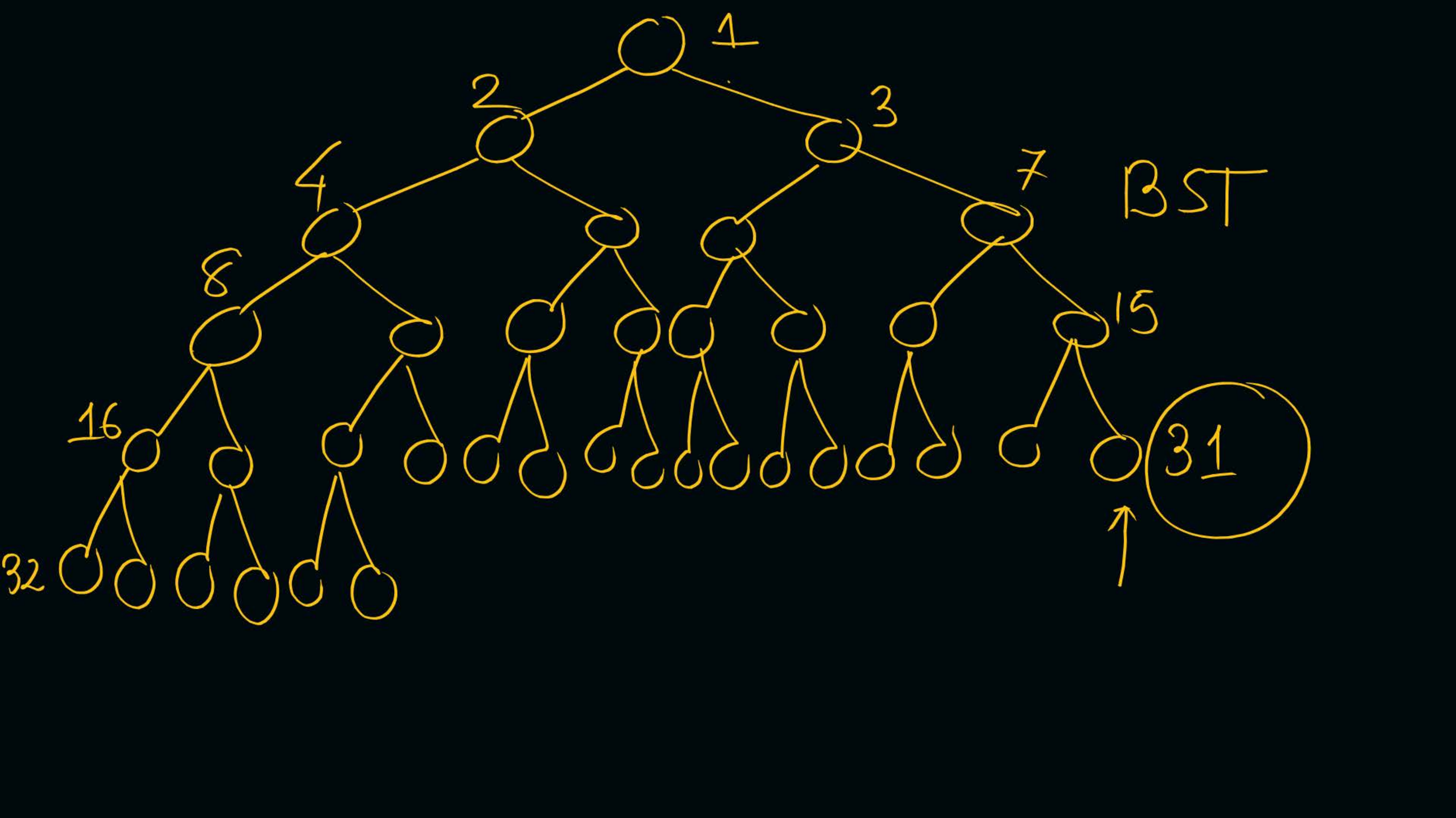


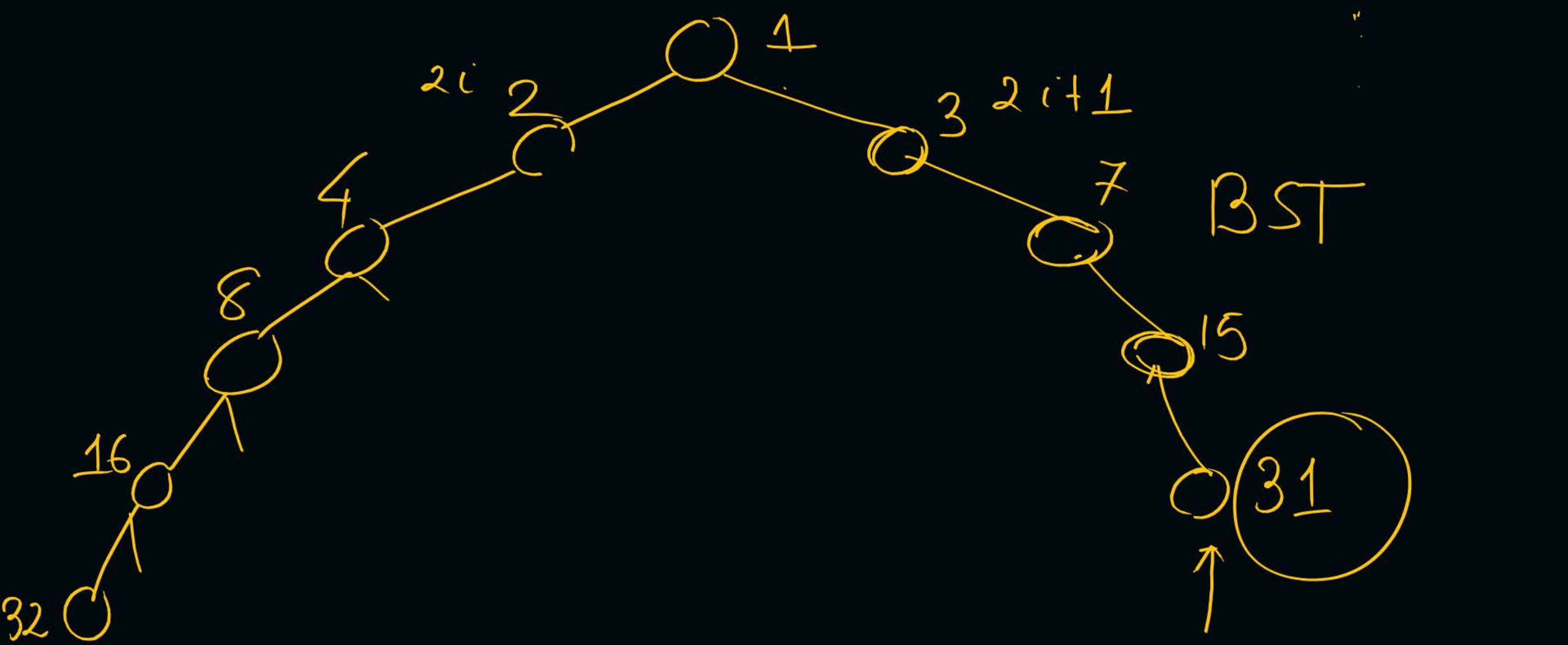
Topic : Heap

Suppose a binary search tree with 37 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with the 1 largest element of the tree is stored at index X and smallest element is stored at Index y then X+Y is_____ .

31 + 32 = 63

Slide


1
2
3
4
7
BST
8
15
16
31
32

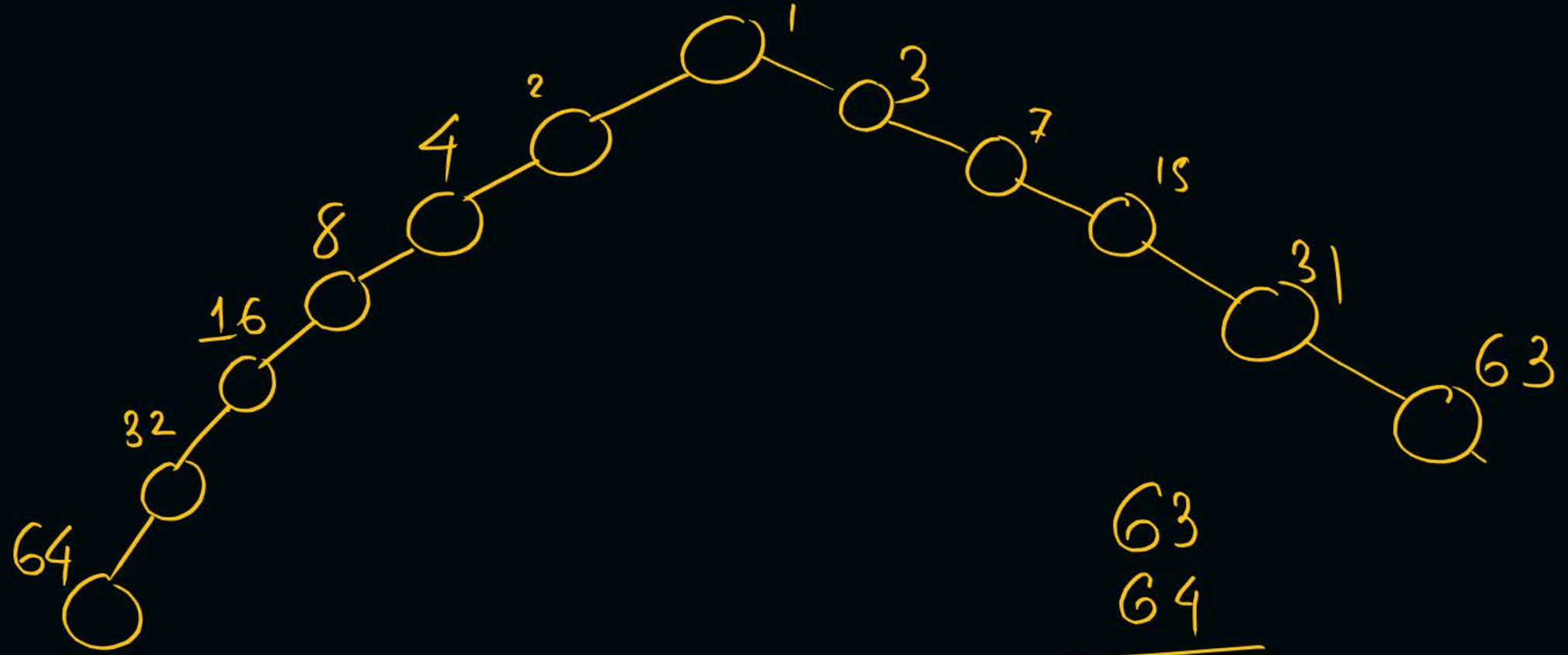

1
2i
2
3
2i+1
4
7
BST
8
15
16
31
32



Topic : Heap

Suppose a binary search tree with 75 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with the 1 largest element of the tree is stored at index X and smallest element is stored at Index y then X+Y is_____ .

Slide



$$
\begin{array}{r}
63 \\
64 \\
\hline
127
\end{array}
$$



Topic : Heap

Heapify : Heapify is a process by which we can Convert any arbitrary tree into Heap (Max, Min)

Min Heap → Max heap / Min

BST → Min Heap / Max

Binay tree - Max / Min

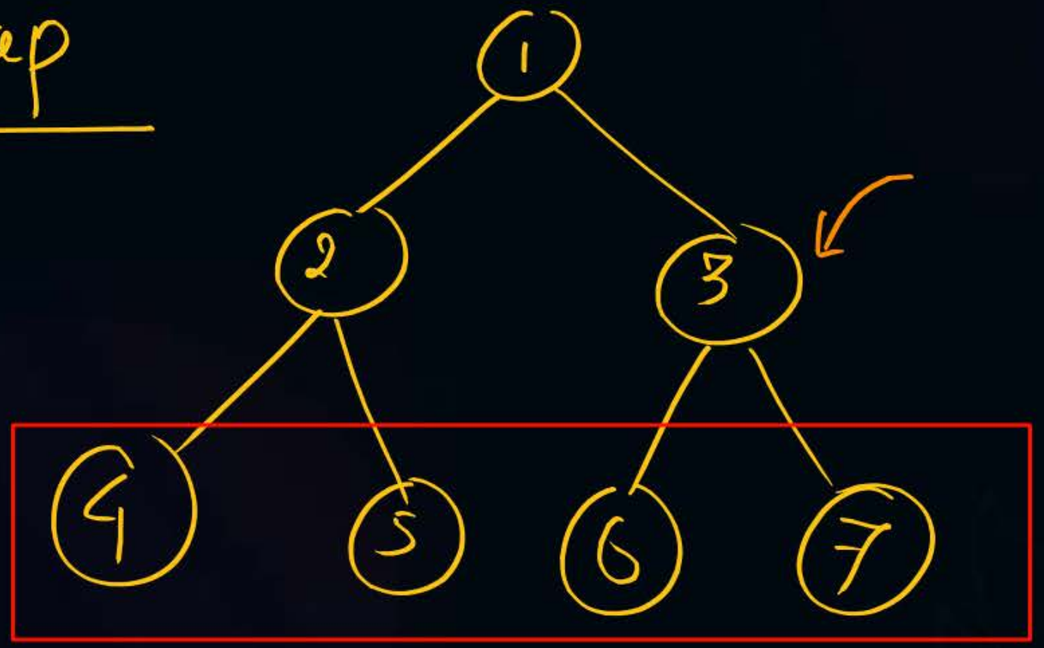
Any tree

Slide



Topic : Heap

Max heap



Heapify

Adjust

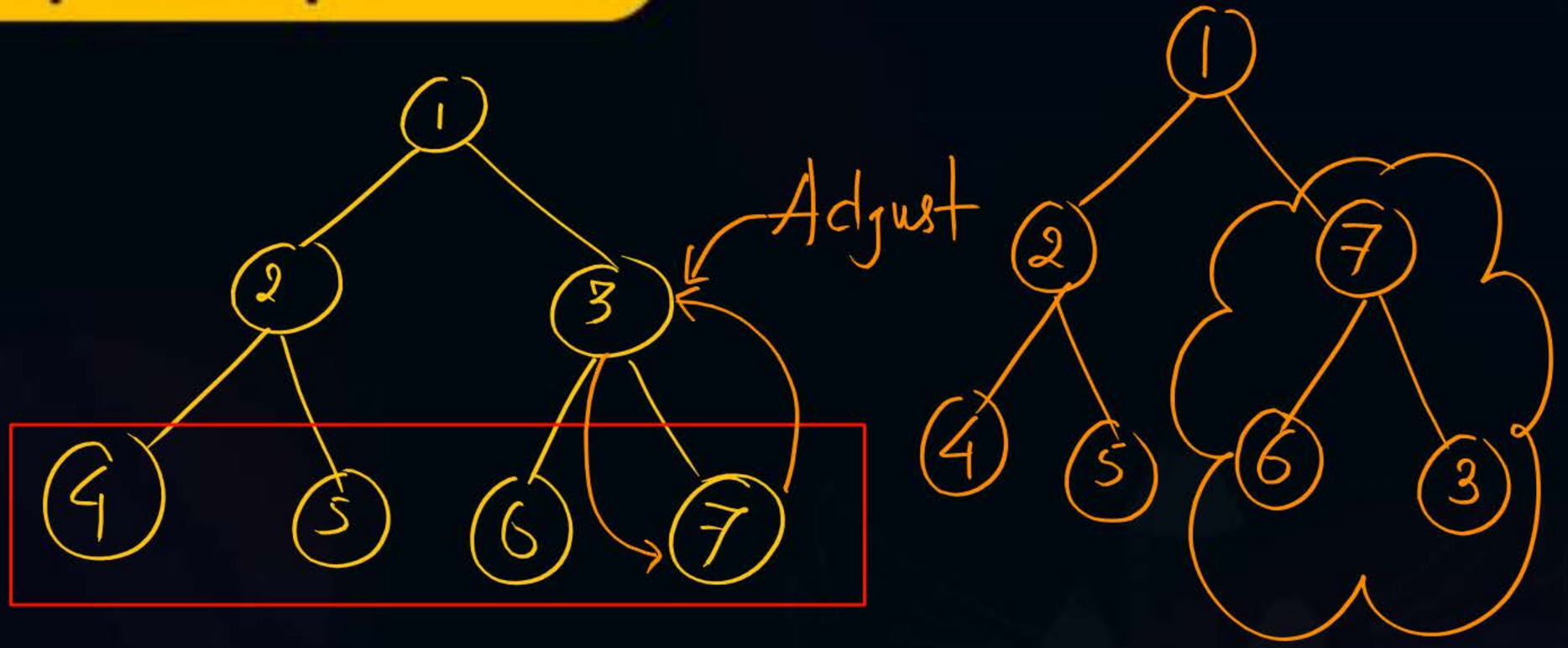
Heapify : Non leaf

Node start

Slide



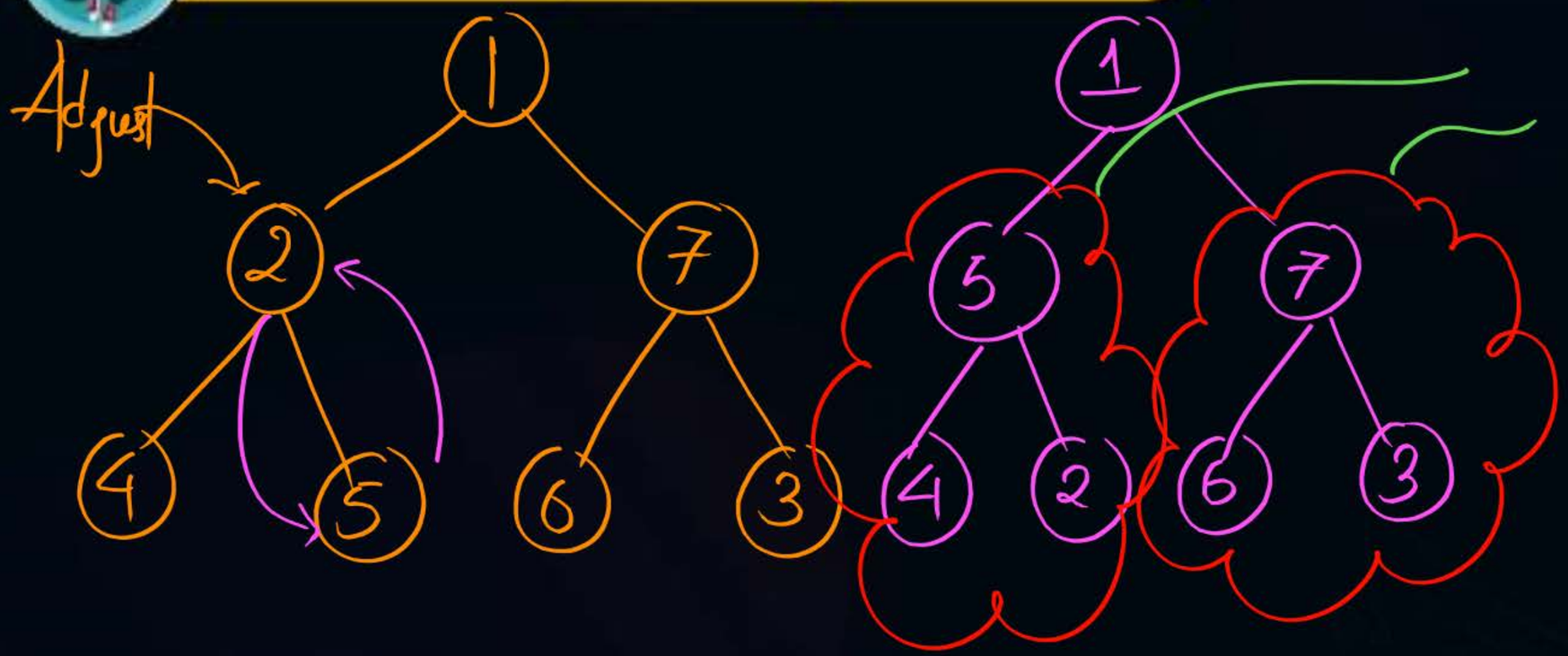
Topic : Heap



Slide



Topic : Heap

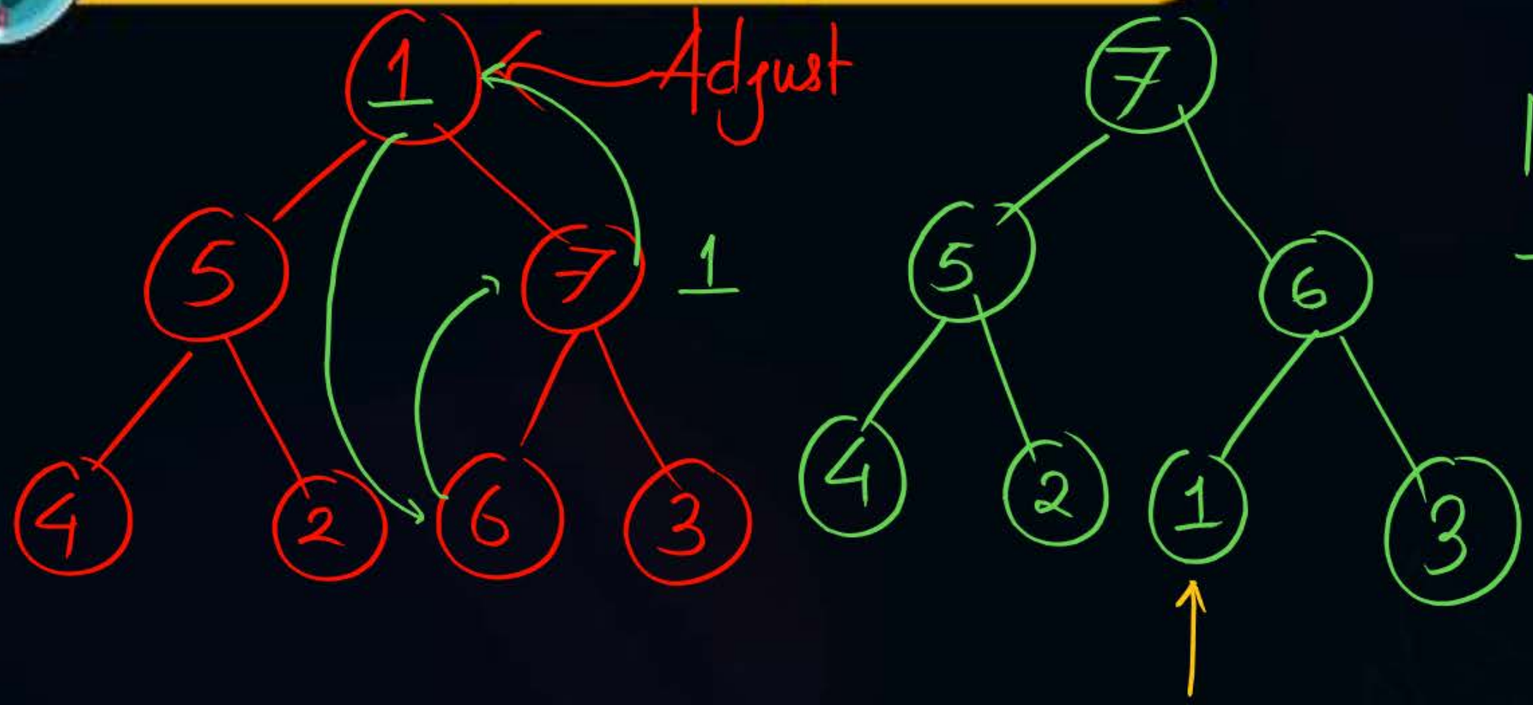




Slide



Topic : Heap



Max heap

Slide



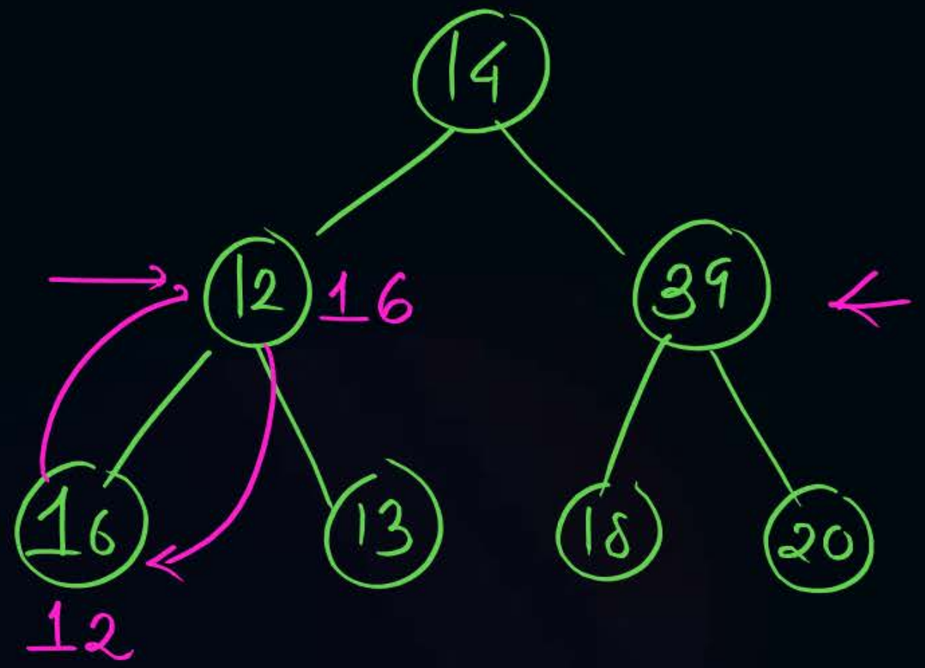
Topic : Heap

if the array given is {14, 12, 39, 16, 13, 18, 20}
is Heapified. The Sum of first three element is ——?
(Max)

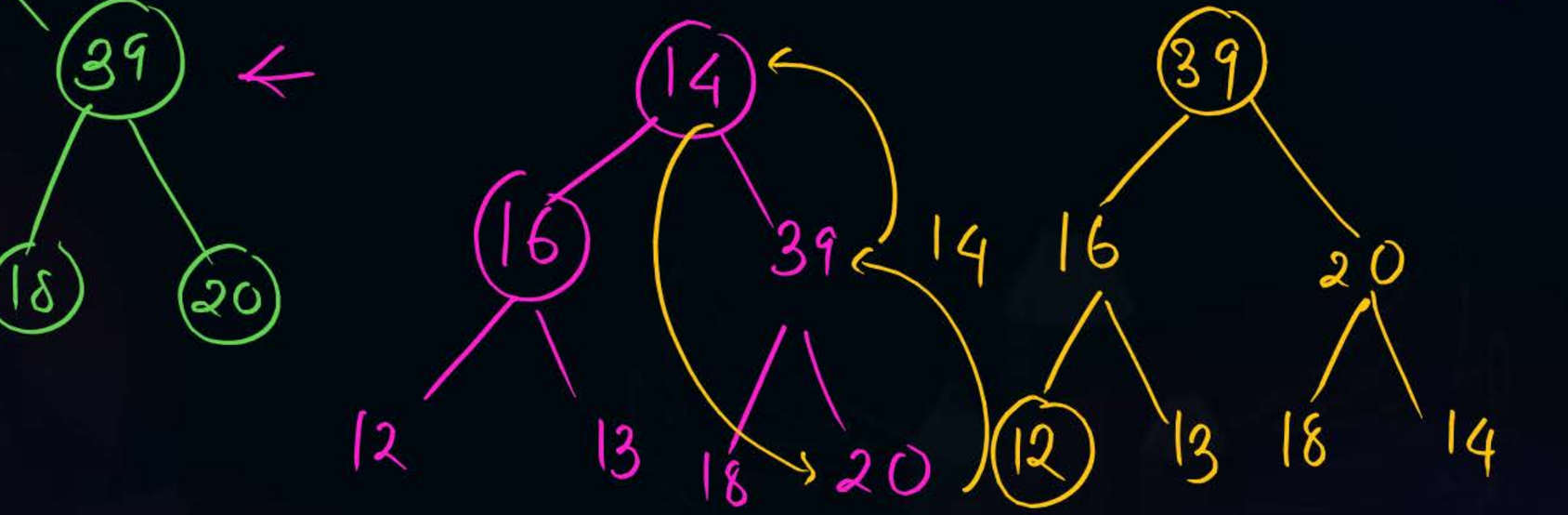
Slide



Topic : Heap



{14, 12, 39, 16, 13, 18, 20}

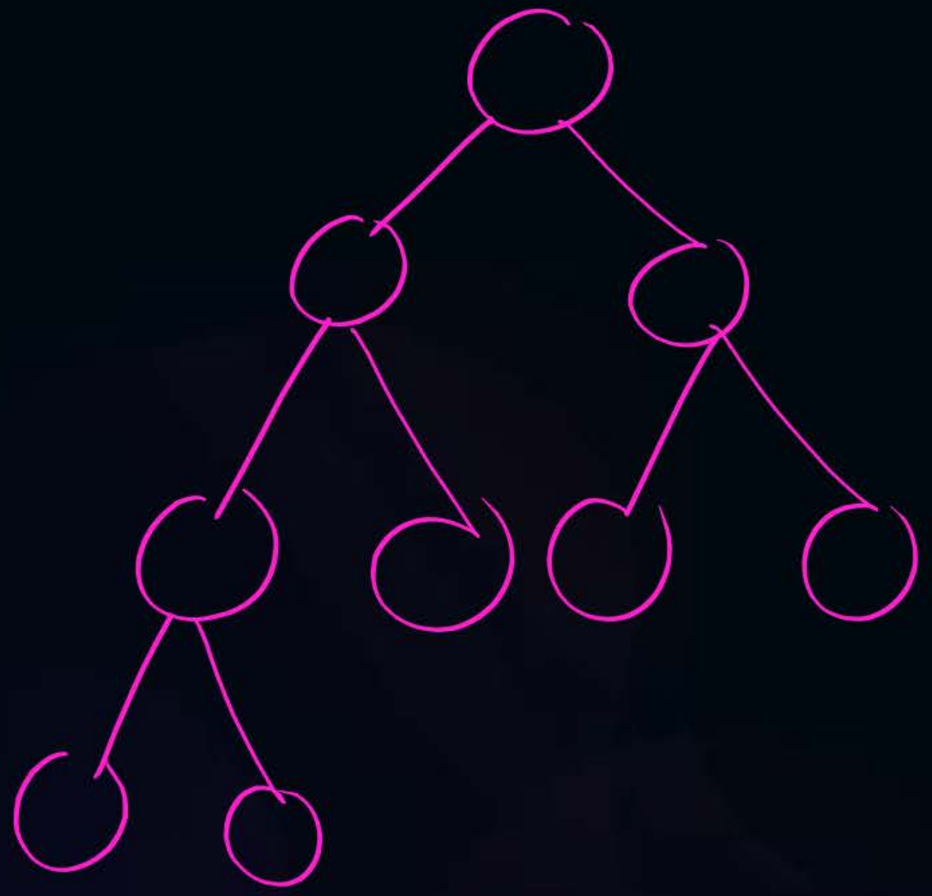


39+20+16 = 75

Slide



Topic : Heap



9

Leaf Node

$\left\lceil \frac{9}{2} \right\rceil = \lceil 4.5 \rceil = 5$

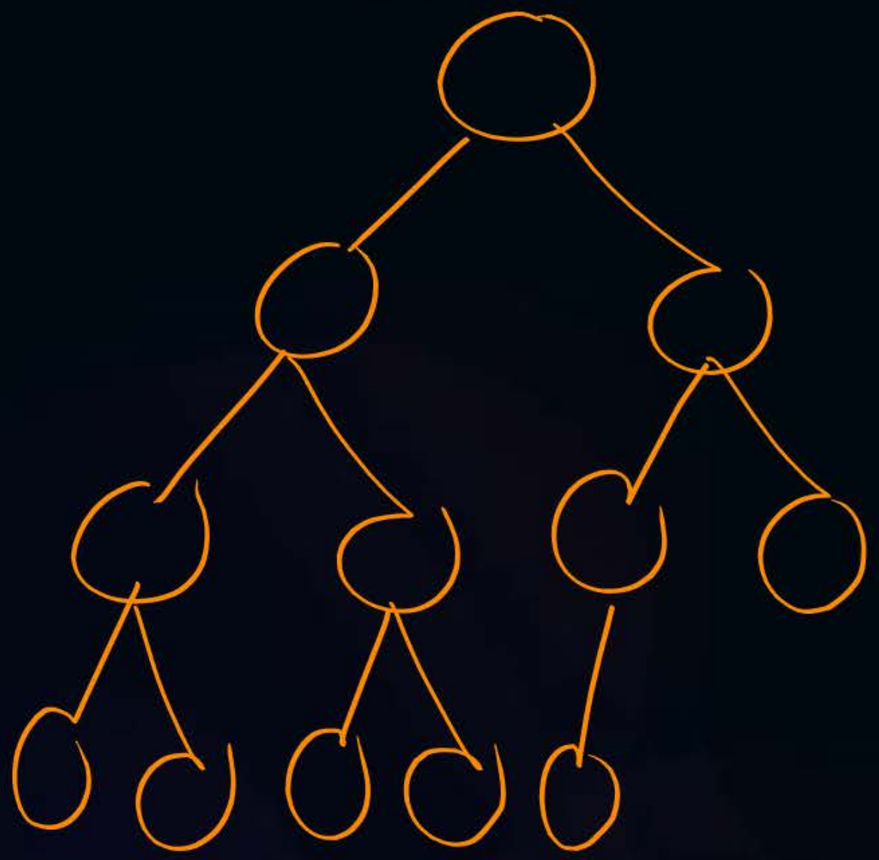
Non Leaf

$\left\lfloor \frac{9}{2} \right\rfloor = \lfloor 4.5 \rfloor = 4$

Slide



Topic : Heap



$$\text{13 - leaf Node} = \left\lceil \frac{n}{2} \right\rceil$$

$$\left\lceil \frac{12}{2} \right\rceil = 6$$

Non leaf Node

$$\left\lfloor \frac{12}{2} \right\rfloor = 6$$

Slide



Topic : Heap

CBT — No. of Leaf Node : $\lceil \frac{n}{2} \rceil$

CBT : No. of Internal Node : $\lfloor n/2 \rfloor$

CBT Height $\lfloor \log_2 n \rfloor$

Slide



Topic : Question

The number of nodes a heap of height k can hold is

(A) 2^k to 2^{k+1}- 1

(B) 2^{k+1} to 2^{k+1}- 1

(C) 2^{k-1} to 2^{k+1}- 1

(D) 2^k -1to 2^{k+1}- 1

Slide



Topic : Heap

Consider a binary min-heap containing 169 distinct elements. Number of leaf present in heap is

$$\left\lceil \frac{169}{2} \right\rceil$$

$$\lceil 84.5 \rceil$$

$$= 85$$

Slide

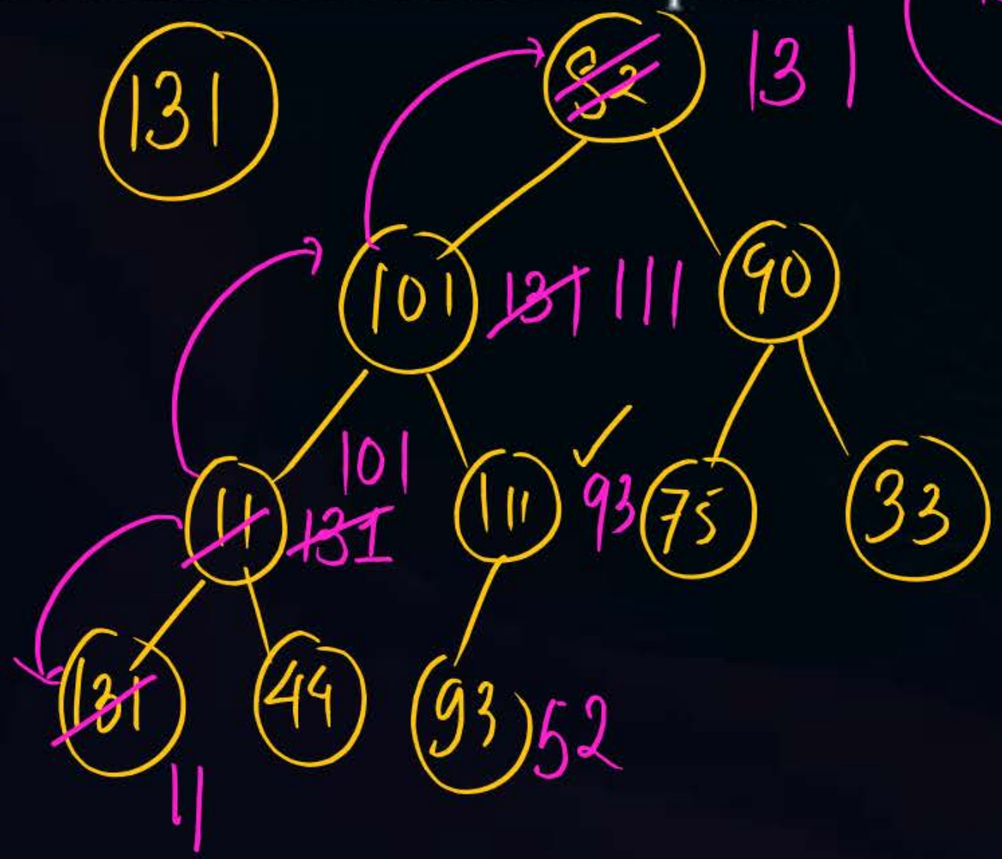
Question: Heapify

2024 Set-2

An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?

(A) 82, 90, 101
(B) 82, 11, 93
(C) 131, 11, 93
(D) 131, 111, 90

131, 111, 90



Slide



Topic : Question

Consider the function and the binary tree shown.

```
typedef struct node {
    int val;
    struct node *left, *right;
} node;
```

HW

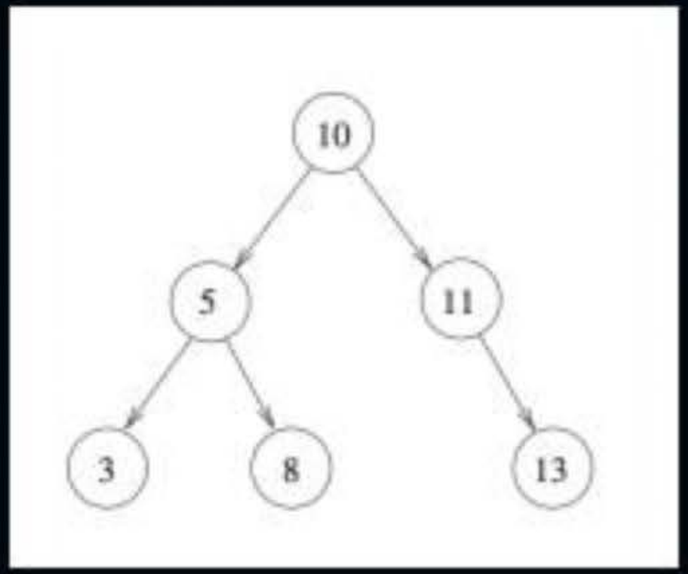
```
int foo(node *p) {
    int retval;
    if (p == NULL)
        return 0;
    else {
        retval = p->val + foo(p->left) + foo(p->right);
        printf("%d ", retval);
        return retval;
    }
}
```

Slide



Topic : Question

Consider the function and the binary tree shown.



HW

```
int foo(node *p) {
    int retval;
    if (p == NULL)
        return 0;
    else {
        retval = p->val + foo(p->left) + foo(p->right);
        printf("%d ", retval);
        return retval;
    }
}
```

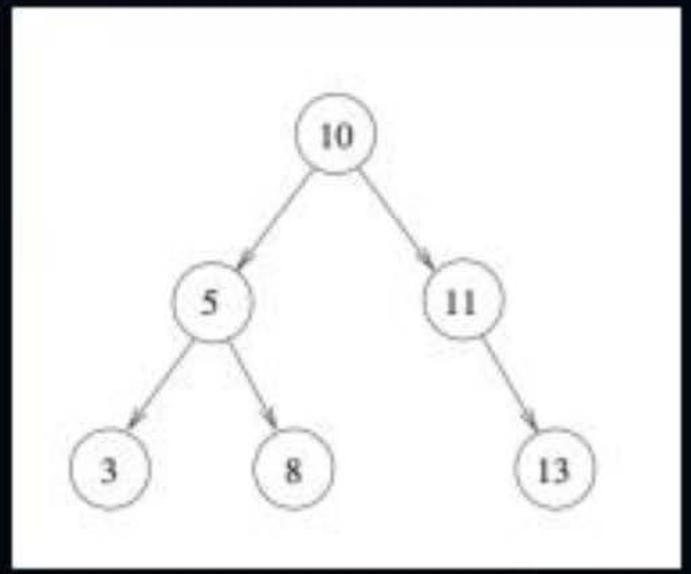
A. 3 8 5 13 11 10
B. 3 5 8 10 11 13
C. 3 8 16 13 24 50
D. 3 16 8 50 24 13

Slide



Topic : Question

Consider the function and the binary tree shown.




```
int foo(node *p) {
    int retval;
    if (p == NULL)
        return 0;
    else {
        retval = p->val + foo(p->left) + foo(p->right);
        printf("%d ", retval);
        return retval;
    }
}
```

HW

Slide



2 mins Summary

Topic — Heap Delete

Topic — Adjust

Topic — Heapify

Topic

Topic

Slide

THANK - YOU